

The Verified-Candidate Loop for Budgeted Large Language Model-Assisted High-Level Synthesis

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Abstract—High-level synthesis (HLS) agents must repair and optimize unfamiliar kernels within limited model and tool budgets. The Verified-Candidate Loop (VCL) lets validation evidence control candidate promotion. Its quality-of-results (QoR) score aggregates effective latency, capacity-normalized resource footprint, and source-change evidence into a single bounded utility. QoR-guided retrieval-augmented generation (QoR-RAG) reuses compatible evidence, and Failure Reflection turns tool diagnostics into the next constraint without extra model calls. We evaluate three open-weight large language model endpoints on a balanced 150-task suite spanning six HLS capabilities. Qwen3.6-27B, the smallest model (27B dense), achieves the highest completion rate (98.7%, 148/150 tasks) and the best estimated optimization scores (79.0 average on QoR tasks). Qwen3.5-122B-A10B (140 completions) is the most token-efficient. DeepSeek V4 Pro (144 completions) consumes 5.87M tokens but returns no measurable QoR improvement. Only explicit optimization tasks surpass the baseline Q_{HW} ; repair tasks achieve correctness without quality gains. Model architecture appears to matter more than parameter count for HLS repair tasks.

Index Terms—high-level synthesis, LLM agent, FPGA, retrieval-augmented generation, quality of results

I. EVIDENCE MUST CONTROL LLM EDITS

Large language model (LLM) agents can edit unfamiliar high-level synthesis (HLS) kernels, but only tool reports certify correctness and hardware quality. A metered agent therefore needs a promotion rule that preserves a verified fallback when any required validation gate fails [1].

Prior HLS agents automate code transformation, directive search, and feedback-driven repair [2]–[4]. Tool-using language agents alternate reasoning and actions [5]. Our design assigns candidate promotion to deterministic HLS evidence under explicit model and tool budgets.

Our Verified-Candidate Loop (VCL) contributes three findings. **(1) Validation contract.** A fail-closed workflow preserves a verified fallback across seven gates: Interface, CSIM, SYNTH, Frequency, Capacity, required CoSIM, and Metric completeness. **(2) Hardware objective.** The bounded score Q_{HW} combines effective latency, capacity-normalized resource footprint, and source-change evidence to reject speedups whose resource cost outweighs their performance gain. **(3) Endpoint evaluation.** Section IV reports cross-model results: Qwen3.6-27B, the smallest model, achieves the highest completion rate (98.7%) and best optimization quality. Only explicit optimization tasks exceed baseline quality; model architecture matters more than parameter count for HLS repair.

II. TOOL EVIDENCE CONTROLS THE AGENT WORKFLOW

The task interface exposes an editable kernel, public tests, target part, and budgets while withholding hidden tests and

frozen references. The controller records every candidate, gate, metric, token, and credit in one run state.

The VCL runs seven gates in order: Interface, CSIM, SYNTH, Frequency, Capacity, required CoSIM, and Metric completeness. Frequency enforces 100 MHz, Capacity enforces U55C limits, and CoSIM applies only when required. Define validity $V(c) = \prod_i G_i$ over applicable gates $G_i \in \{0, 1\}$; an inapplicable gate contributes 1. Let b be the verified fallback. The promotion rule is $P(c) = V(c) \wedge [Q_{HW}(c) > Q_{HW}(b)]$. Budget admission reserves credits for required gates; denial or failure retains b . QoR-RAG supplies version-compatible rules and measured outcomes. Failure Reflection converts diagnostics into the next constraint without another LLM request. Duplicate candidates and repeated failures stop retries.

Figure 1 summarizes the loop. The three campaigns record 691 candidate checks, 112 interface rejections, 487 synthesis gates, and 67 CoSIM gates, establishing contract execution without isolating mechanism effects.

III. ONE SCORE REJECTS FALSE SPEEDUPS

The hardware quality score Q_{HW} combines performance gain and resource cost into a single bounded utility. Let anchor a and candidate c have cycle latency L , effective period p , and initiation interval II . The performance ratio $r_P = (p_a L_a)/(p_c L_c)$ captures speedup weighted by pipeline overhead. The area ratio r_A uses a capacity-normalized footprint $F(x) = \sum_r x_r / K_r$ over LUT, FF, DSP, BRAM, and URAM to prevent a resource transfer from causing a discontinuous jump. Two evidence factors reward any source edit ($D, 1.01\times$) and validity repair ($F, 2\times$). The aggregate hardware ratio $h = 1.01^D 2^F r_P^{0.55} r_A^{0.45}$ weights performance at 55% and area at 45%, calibrated against 99 frozen PPA references from Vitis 2025.2.

The bounded utility $Q_{HW} = 1 - 1/(1+h)^2$ maps $h=1$ (no change) to 0.75, $h \rightarrow \infty$ to 1, and $h \rightarrow 0$ to 0, with continuous monotonic degradation for bloated speedups. Budget efficiency $E = \max(0.80, 1 - 0.10u_{\text{credit}} - 0.10u_{\text{time}})$ discounts excessive tool calls with a floor at 0.80, where each u term is the consumed fraction clipped to $[0, 1]$.

The final score is

$$S = 100 V(c) Q_{HW} E. \quad (1)$$

Promotion compares Q_{HW} only after $V(c)=1$; remaining budget never promotes an inferior design. A failed gate yields $S=0$; the efficiency floor ensures a valid baseline earns at least 60 points.

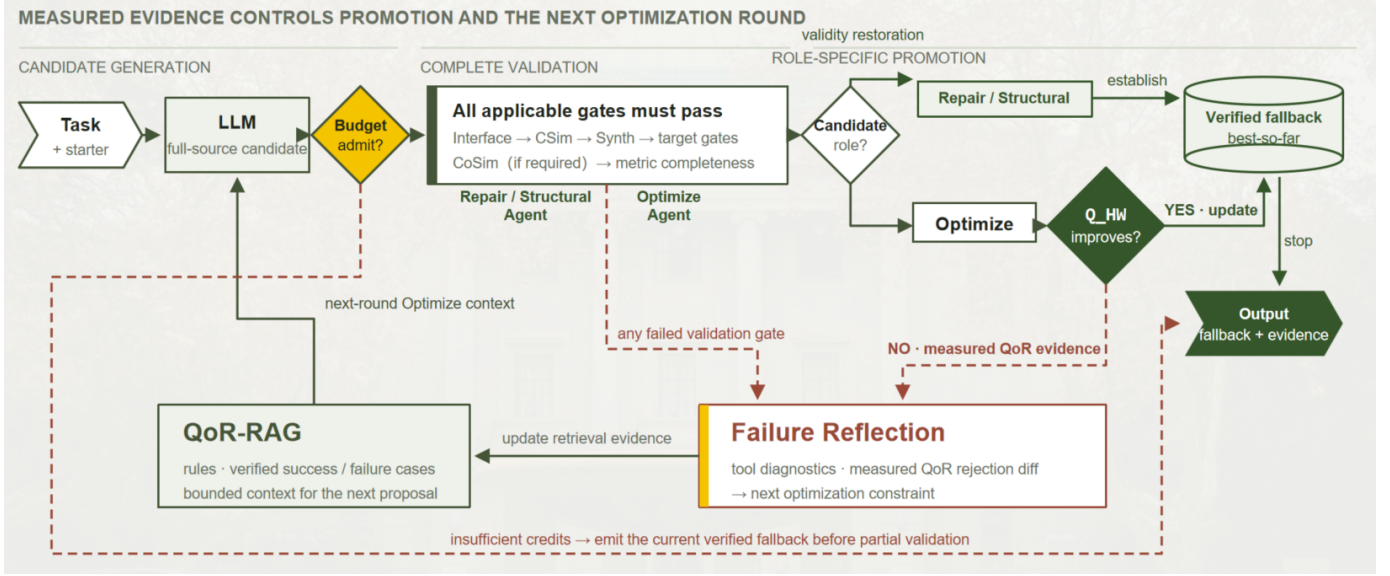


Fig. 1. **The Verified-Candidate Loop.** A candidate replaces the verified fallback only after every applicable gate passes and Q_{HW} improves. Failure Reflection converts tool diagnostics into the next constraint without extra model calls.

IV. COMPLETION OUTPACES SCORE COVERAGE

Protocol. Three hosted endpoints process the same balanced 150-task suite under one agent revision. Each completed output must pass CSIM, SYNTH, 100 MHz, U55C capacity, and applicable CoSIM under the Track-A contract [6]. The models span three architecture-scale regimes: DeepSeek V4 Pro (1.6 T MoE, 49 B active), Qwen3.5-122B-A10B (122 B MoE, 10 B active), and Qwen3.6-27B (27 B dense).

TABLE I

THE SMALLEST DENSE MODEL LEADS. QoR SCORES ARE ESTIMATED AS 100 Q_{HW} E ; DEEPSEEK RETURNED NO MEASURABLE Q_{HW} IMPROVEMENT.

Endpoint	Completed	Tokens (M)	Credits	Est. QoR mean	>76 rate
DeepSeek V4 Pro	144/150 (96.0%)	5.87	2,617	—	—
Qwen3.5-122B-A10B	140/150 (93.3%)	1.68	2,375	74.0	7/140 (5%)
Qwen3.6-27B	148/150 (98.7%)	1.92	2,515	74.6	15/148 (10%)

Results. Table I summarises the three campaigns. Qwen3.6-27B achieves 98.7% completion (148/150 tasks, 2 failures), including 25/25 in the challenging structural CoSIM repair category. Despite being the smallest model, it outperforms both larger MoE models on completion and optimization quality. Qwen3.5-122B-A10B is the most token-efficient (1.68M tokens), while DeepSeek V4 Pro consumes 5.87M tokens (3.1× more than Qwen3.6) with no measurable QoR improvement.

Per-category breakdown (Appendix C) reveals four patterns. (1) Only explicit QoR optimization tasks yield scores above baseline; all five repair/generation categories cluster at $Q_{HW} \approx 0.7576$ ($S \approx 73.7$ with $E = 0.973$). The agent passes validation gates but performs no architecture-level optimization on repair tasks. (2) Qwen3.6-27B dominates on optimization quality (avg. $Q_{HW} = 0.812$, 60% of optimization tasks >76, 52% >80), while Qwen3.5 achieves avg. $Q_{HW} = 0.775$ (28%

>76). (3) Model architecture appears more predictive of HLS repair success than parameter count: the 27 B dense model outperforms both 122 B and 1.6 T MoE models on structural repair (100% vs. 80% and 92%).

Takeaway. Qwen3.6-27B leads completion (98.7%) and optimization quality (79.0 average QoR score on optimization tasks), suggesting architecture matters more than parameter count for HLS repair. Only explicit optimization tasks benefit from LLM-driven QoR improvement; repair tasks achieve correctness without quality gains.

V. LIMITS AND NEXT STEPS

The VCL makes validation evidence and Q_{HW} the promotion authority. The capacity-normalized resource footprint replaces worst-single-growth with a continuous aggregate ratio, and D/F evidence factors reward source change and validity repair. Qwen3.6-27B, the smallest model tested, leads both completion (98.7%) and optimization quality (79.0 average on QoR tasks). Only explicit QoR optimization tasks improve beyond the $Q_{HW} \approx 0.7576$ baseline.

The agent depends on mature testbench inputs, a fixed toolchain (Vitis 2025.2), and a single FPGA platform (Alveo U55C). Matched ablations are required to isolate causal mechanisms: full agent, no QoR-RAG, no measured-failure retrieval, and no Failure Reflection. Each variant must report campaign identifier, success, repeated failures, mean Q_{HW} , tokens, calls, credits, and wall time. Broader corpora and cross-platform transfer remain for future work.

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APPENDIX A BENCHMARK COMPOSITION

The manifest freezes every upstream path, commit, task description, test, and gate configuration, retaining 150 tasks with no cross-category kernel overlap. Table II describes the six balanced capabilities, each containing 25 tasks. Only the structural category requires C/RTL co-simulation.

TABLE II
TASK SUITE COMPOSITION.

Task type	Tasks	CoSim
Code generation (stub $\rightarrow$ kernel)	25	0
Compile repair	25	0
Synthesis repair	25	0
Functional repair (CSIM)	25	0
Structural repair (CoSim)	25	25
QoR optimization	25	0

Kernels come from two public AMD repositories at fixed commits: Xilinx/Vitis-HLS-Introductory-Examples (109 tasks, Apache-2.0, aa5c160f) and Xilinx/Vitis_Accel_Examples (41 tasks, MIT, 81187602). The 150 variants reuse 65 unique source paths.

APPENDIX B TRACK-A REQUIREMENT MAPPING

All tasks require CSIM and SYNTH; the manifest requires CoSim for the 25 structural tasks only. Table III maps Track-A requirements to recorded evidence.

TABLE III
TRACK-A RUN CONTRACT.

Requirement	Recorded evidence
U55C; Vitis 2025.2	Container pins target part and tool release; reports preserve execution-source revision.
CSIM, SYNTH, applicable	Missing or failed gate rejects candidate. Hidden tests and references remain evaluator-only.
CoSim	
≥ 100 MHz; feasible	Frequency and U55C capacity gates precede promotion. Reports retain latency, II , period, LUT, FF, DSP, BRAM, URAM.
Token/credit accounting	Each run records tokens and credits. Limit: 9,000 credits.
3-endpoint comparison	DeepSeek V4 Pro, Qwen3.5-122B-A10B, Qwen3.6-27B; one manifest, one source revision.
Docker; reproducibility	Docker workflow, public task materials, structured reports, result-generation script provided.
PDF, source, video	Authors confirm IEEE format, 2-page boundary, archive contents, and 5-minute demo.

The three checkpoints span three architecture regimes: DeepSeek V4 Pro (1.6 T MoE, 49 B active, FP8), Qwen3.5-122B-A10B (122 B MoE, 10 B active, AWQ-4bit), and Qwen3.6-27B (27 B dense, FP8).

APPENDIX C DETAILED THREE-ENDPOINT RESULTS

Tables IV and V provide the per-category breakdown underlying Section IV. Scores are 100 $Q_{HW} E$ (Eq. (1)).

TABLE IV
PER-CATEGORY COMPLETION BY ENDPOINT. EACH CATEGORY: 25 TASKS. QoR MEAN: OPTIMIZATION TASKS ONLY.

Category	DeepSeek V4	Qwen3.5-122B	Qwen3.6-27B
Code generation	22/25 (88%)	22/25 (88%)	24/25 (96%)
Compile repair	25/25 (100%)	25/25 (100%)	25/25 (100%)
Synthesis repair	25/25 (100%)	24/25 (96%)	25/25 (100%)
Functional repair	24/25 (96%)	24/25 (96%)	24/25 (96%)
Structural repair	23/25 (92%)	20/25 (80%)	25/25 (100%)
QoR optimization	25/25 (100%)	25/25 (100%)	25/25 (100%)
QoR mean (opt.)	—	75.4	79.0

TABLE V
TOKEN CONSUMPTION AND CREDIT EXPENDITURE. DEEPSEEK USED 3.1 $\times$ MORE TOKENS THAN QWEN3.6 WITH NO QoR GAIN.

Endpoint	Tokens (M)	Credits
DeepSeek V4 Pro	5.87	2,617
Qwen3.5-122B-A10B	1.68	2,375
Qwen3.6-27B	1.92	2,515

APPENDIX D REPRODUCTION

Task list: tasks/track_a_150/candidate_manifest.json. Campaigns: runs/150_ultimate/. Run `python3 technical-paper/scripts/update_results.py` from the repository root to refresh macros; the script validates model identity, task count, API client, execution-source hash, and credit limit before emitting output.

All campaigns: source snapshot 0a06af39777b6ae7f3962afa2910232eaf782e91727e0f184ec168f1a41f106c, temperature 0, 4,096-token output limit, 180 s request timeout, ≤ 2 retries.